Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Architect · Agentic AI · Integration Architecture · Context Engineering · HITL · Enterprise Solution Delivery · GenAI

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Virginia, USA · F-1 STEM OPT → H-1B

PROFESSIONAL SUMMARY

8+ years architecting production AI systems and agentic workflows, with a track record of cross-functional solution delivery across compliance, operations, and product stakeholders. Deployed an agentic AI system for marketing, real estate, and commercial survey clients — client-facing scope, production guardrails, FAA compliance enforcement. Built **Leasel** — an enterprise integration architecture: document ingestion → structured extraction → rules-based eligibility → 8-agent LangGraph StateGraph → human-review gate; designed for regulated environments where LLM reasons but deterministic logic enforces every compliance call. At Entrata, defined integration architecture across 20+ service boundaries and data pipelines for a 100K+ MAU platform — translating federal housing regulations into production-ready engineering specs across 120+ Agile sprints. Core principle: *ethical AI is an architecture decision enforced in code, not a policy document.* M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI Architecture | Agent Workflow Integration | Integration Architecture | Context Engineering | Contextual Knowledge Mapping | LangGraph Multi-Agent | HITL Design | Ethical AI & Guardrails | GenAI / Prompt Engineering | Enterprise Data Pipelines | Production AI Solution Delivery | Cross-Functional Stakeholder Delivery

EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer — Client-Facing Agentic Deployment · Multi-Stakeholder GenAI · Compliance Guardrails

- Deployed an agentic AI system serving **marketing, real estate, and commercial survey clients** — client-facing scope across three distinct verticals from one architecture: parsed natural-language survey requests into structured flight plans with photogrammetry math (ground sampling distance, camera footprint, overlap-driven line spacing).
- Architected a **LangGraph stateful agent workflow** (parse → plan → retrieve → validate → conditional routing) with compliance enforcement gates: compliant missions route to MAVLink/QGroundControl export, non-compliant ones route to rejection with grounded regulatory citations.
- Implemented **context engineering** via a RAG layer over FAA Part 107 and operational-guidance corpus — every compliance check functions as a hypothesis test cited to source regulation; designed backend-agnostic retriever with ChromaDB/embedding upgrade path.
- Engineered **ethical AI guardrails** enforcing FAA altitude limits, battery budgets, and geometric no-fly-zone checks deterministically — blocking non-compliant actions before any agent export fires.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Quantitative Research · Anomaly Detection · Teaching (600+ students)

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines, surfacing data integrity issues manual review missed.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Integration Architecture · Enterprise Data Pipelines · Production AI · Cross-Functional Delivery

- **Defined integration architecture across 20+ service boundaries** and data pipelines for a 100K+ MAU platform — led microservices redesign (Docker, 50% scalability gain, 30% deployment reduction), converting 20% of product modules to service-based architecture with reduced interdependencies.
- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated data processing workflows replacing manual Excel operations; output validation, quality gates, and compliance controls enforced at launch.
- **Cross-functional solution delivery:** collaborated across compliance, operations, product, and finance stakeholders across 120+ Agile sprints — translated federal housing regulations (HAP, AMI, HUD, LIHTC) into production-ready engineering specs; zero compliance violations.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines (100% of resident SSNs/phones encrypted); zero security incidents; updated 50%+ of product codebase for encryption compliance.
- Optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ junior engineers across 300+ peer code reviews.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · AWS S3

- Built full-stack modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts processing 1.6M records; migrated thousands of client documents to AWS S3 and redesigned document storage database architecture.

PORTFOLIO PROJECTS

Leasel — Enterprise AI Integration Architecture with HITL & Ethical Guardrails

Agentic HITL · Integration Architecture · Ethical AI

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a full **enterprise integration architecture**: document ingestion → structured extraction (OCR-fallback) → context-aware chunking → TF-IDF contextual knowledge retrieval → 8-agent LangGraph StateGraph → rules-based eligibility gate → human-review escalation; designed for regulated environments.
- Designed **8-agent agent workflow** (Triage → Retrieval → Config → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with real conditional edges and confidence-gated HITL — correctly traces false-ineligibility ticket to misconfigured setting + release tag end-to-end.
- Core **ethical AI design rule enforced in code**: LLM extracts and reasons; deterministic logic or human review makes every final regulated call — 100% citation correctness, 0% unsupported-claim rate, 0 safety boundary violations on eval set.

AgentProbe — Agentic AI Dev Platform with CI-Integrated Evaluation (SDLC)

GenAI Platform · LLM Eval · SDLC

2024–Present · Python, LangGraph, LangChain, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **SDLC-integrated agentic evaluation pipeline**: 5-metric LLM-as-judge scoring (factual accuracy, hallucination, retrieval, completeness, compliance) runs on every commit via GitHub Actions — regressions blocked before merge.
- Deployed a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with multi-provider routing (Anthropic, OpenAI, Groq, Ollama), tool-use patterns, inter-agent state handoffs, and 15-endpoint FastAPI backend.
- Delivered **trace-level observability** across concurrent agent test suites — per-run metrics, regression drift, retrieval quality, and latency — via React dashboard with ChromaDB RAG knowledge pipeline.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Agentic AI & Architecture: LangGraph, LangChain, Agentic AI, Agent Workflow Integration, Multi-Agent Orchestration, HITL, Context Engineering, Contextual Knowledge Mapping, Spec Engineering, Ethical AI, Integration Architecture, Enterprise Architecture, Solution Delivery

GenAI & LLM: GenAI, Prompt Engineering, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Production AI, Foundation Model APIs, Multi-Provider LLM Routing, ChromaDB, Embeddings, OpenAI API, Anthropic API, Groq, Ollama

Backend & Infrastructure: Python, FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), SDLC, PostgreSQL, MySQL, Microservices, Data Pipelines, ETL, RabbitMQ, Libsodium, Linux

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, NLP, spaCy, BERT, NumPy, Pandas, Statistical Modeling, Feature Engineering

Languages: Python (primary), PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash